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Optical biosensor device with optical signal enhancement structure

Abstract

The present disclosure relates to an integrated chip including a semiconductor layer and a photodetector disposed along the semiconductor layer. A color filter is over the photodetector. A micro-lens is over the color filter. A dielectric structure comprising one or more dielectric layers is over the micro-lens. A receptor layer is over the dielectric structure. An optical signal enhancement structure is disposed along the dielectric structure and between the receptor layer and the micro-lens.

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Background/Summary

BACKGROUND

(1) In recent years, the semiconductor industry has developed integrated chips (ICs) having integrated bio-sensors configured to detect the presence of certain bio-markers in a sample solution (e.g., in a patient's blood). Bio-sensors are analytical devices that convert a biological response into an electrical signal. For example, bio-sensors can generate electrical signals that identify and detect different analytes such as toxins, hormones, DNA strands, proteins, bacteria, etc., in a variety of applications such as molecular diagnostics, pathogen detection, and environmental monitoring. The integration of bio-sensors in system-on-chips (SOCs) provides for promising avenues in the development of diagnostic tools for infectious diseases and cancers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. **1-5** illustrate cross-sectional views of some embodiments of an optical biosensor integrated chip including a receptor layer disposed directly over a photodetector and an optical signal enhancement structure disposed between the receptor layer and the photodetector.

(3) FIG. **6** illustrates a cross-sectional view of some embodiments of an optical biosensor integrated chip including a receptor layer disposed directly over a plurality of photodetectors and an optical signal enhancement structure disposed between the receptor layer and the plurality of photodetectors.

(4) FIGS. **7**, **8**A, **8**B, **9**, and **10** illustrate top views of some embodiments of the optical biosensor integrated chip of FIG. **6**.

(5) FIGS. **11-39** illustrate cross-sectional views of some embodiments of a method for forming an optical biosensor integrated chip including a receptor layer disposed directly over a photodetector and an optical signal enhancement structure disposed between the receptor layer and the photodetector.

(6) FIG. **40** illustrates a flow diagram of some embodiments of a method for forming an optical biosensor integrated chip.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) Optical biosensor devices include elements for detecting an analyte using optical radiation. For example, some optical biosensors include an optical radiation source (e.g., a laser light source), an optical apparatus (e.g., including a lens, a mirror, an optical prism, or the like), a bioreaction device (e.g., a biosensing interface having bioreceptors disposed thereon), and an image sensor (e.g., a photodetector or the like). In some optical biosensors, each of the optical radiation source, the optical apparatus, the bioreaction device, and the image sensor are separate devices assembled together as part of a larger apparatus. A challenge with these optical biosensor devices is that they are typically large, bulky, and/or unportable.

(10) Various embodiments of the present disclosure are related to an optical biosensor device comprising a bioreaction device integrated with an image sensor for improving a portability of the

optical biosensor device. For example, the optical biosensor device includes a photodetector disposed along the semiconductor layer. A color filter is disposed over the photodetector. A micro-lens is disposed over the color filter. A dielectric structure is disposed over the micro-lens. A receptor layer is disposed over the dielectric structure. The receptor layer is configured to receive bioreceptors. An optical signal enhancement structure is disposed along the dielectric structure and between the receptor layer and the micro-lens. During operation, a sample solution comprising analytes is provided to bioreceptors which are immobilized along the receptor layer. The bioreceptors are configured to immobilize the analytes. Further, an optical radiation source is configured to excite a bioreaction between the bioreceptors and the analytes with incident optical radiation. The photodetector is configured to detect a sensor optical radiation signal that is emitted from the bioreaction in response to the excitation.

(11) By integrating the bioreaction device and the image sensor into an integrated chip having a small form factor, the size of the optical biosensor device may be reduced and the portability of the optical biosensor device may be improved. Further, by including the optical signal enhancement structure between the receptor layer and the micro-lens, a performance of the optical biosensor device may be improved. For example, the optical signal enhancement structure is configured to enhance the sensor optical radiation signal before the sensor optical radiation signal reaches the photodetector. Thus, a performance (e.g., a sensitivity, an accuracy, etc.) of the optical biosensor device may be improved.

(12) FIG. **1** illustrates a cross-sectional view **100** of some embodiments of an optical biosensor integrated chip including a receptor layer **118** disposed directly over a photodetector **104** and an optical signal enhancement structure **126** disposed between the receptor layer **118** and the photodetector **104**.

(13) The photodetector **104** is disposed within a semiconductor layer **102**. The semiconductor layer **102** has a frontside **102***a* and a backside **102***b*, opposite the frontside **102***a*. An interlayer dielectric (ILD) structure **108** comprising one or more ILD layers is disposed on the backside **102***b* of the semiconductor layer **102**. An interconnect structure **110** comprising one or more conductive features (e.g., metal lines, vias, contacts, bond pads, etc.) is disposed within the ILD structure **108**. In some embodiments, one or more of the conductive features are coupled to the photodetector **104**.

(14) A color filter **106** is disposed on the frontside **102***a* of the semiconductor layer **102**. A micro-lens **112** is disposed over the color filter **106**. A glue layer **114** is disposed over the micro-lens **112**. A dielectric structure **116** is disposed over the glue layer **114**. In some embodiments, the glue layer **114** is directly over the micro-lens **112** and directly between the micro-lens **112** and the dielectric structure **116**. In some other embodiments, the glue layer **114** is disposed on opposite sides of the color filter **106** and the micro-lens **112** and directly between the semiconductor layer **102** and the dielectric structure **116**, but not directly over the micro-lens **112**. In some embodiments, a cavity (not shown) comprising air or the like exists directly over the micro-lens **112** and directly between the micro-lens **112** and the dielectric structure **116**. In some embodiments, some other suitable optimal material can be disposed directly over the micro-lens **112** and directly between the micro-lens **112** and the dielectric structure **116**. The receptor layer **118** and a capping layer **122** are disposed over the dielectric structure **116**. A microfluidic channel **124** exists within the capping layer **122**. The microfluidic channel **124** is delimited by sidewalls and a lower surface of the capping layer **122**. The capping layer **122** has an inlet **122***a* and an outlet **122***b* to the microfluidic channel **124**. The receptor layer **118** is disposed along the microfluidic channel **124** between the sidewalls of the capping layer **122** and below the lower surface of the capping layer **122** that delimit the microfluidic channel **124**. In some embodiments, an upper surface of the receptor layer **118** further delimits the microfluidic channel **124**.

(15) In some embodiments, the receptor layer **118** is configured to receive a bioreceptor **120** (e.g., an antibody, an enzyme, a nucleic acid, DNA, etc.). For example, in some embodiments, the receptor layer **118** is configured to immobilize a bioreceptor **120**. In some embodiments, a

bioreceptor **120** is disposed (e.g., immobilized) along a top of the receptor layer **118**. The bioreceptor **120** is configured to receive an analyte **132**. For example, the bioreceptor **120** is configured to immobilize an analyte **132**. In some instances, a bioreaction occurs when an analyte **132** is immobilized on a bioreceptor **120**.

(16) In some embodiments, during operation of the optical biosensor integrated chip, a sample solution **130** comprising one or more analytes **132** is provided to the microfluidic channel **124** via the inlet **122***a*. An optical radiation source **134** (e.g., a laser light source or some other optical radiation source) emits incident optical radiation **136** toward the optical biosensor integrated chip. In some instances, when an analyte **132** is immobilized on a bioreceptor **120** and the incident optical radiation **136** impinges on the analyte **132** and/or the bioreceptor **120**, the bioreaction between the analyte **132** and the bioreceptor **120** emits a sensor optical radiation signal **138**. The sensor optical radiation signal **138** passes through the dielectric structure **116**, the glue layer **114**, the micro-lens **112**, and the color filter **106** and impinges on the photodetector **104**. The photodetector **104** is configured to detect the sensor optical radiation signal **138**. The analyte **132** can be determined based on the detected sensor optical radiation signal **138**. For example, a signal processor (not shown) can be coupled to the photodetector **104** and can determine the analyte **132** based on the output of the photodetector **104**.

(17) Because the optical biosensor comprises a bioreaction device (e.g., the receptor layer **118** and/or the bioreceptor **120**) and an image sensor (e.g., the photodetector **104**) implemented in an integrated chip having a small form factor, a portability of the biosensor may be improved.

(18) Further, the biosensor integrated chip includes the optical signal enhancement structure **126** disposed between the receptor layer **118** and the photodetector **104**. The optical signal enhancement structure **126** is disposed along the dielectric structure **116** and between the receptor layer **118** and the micro-lens **112**. The optical signal enhancement structure **126** comprises one or more elements configured to enhance the sensor optical radiation signal **138** before the sensor optical radiation signal **138** reaches the photodetector **104**. For example, in some embodiments, the optical signal enhancement structure **126** is configured to increase an intensity of the sensor optical radiation signal **138** and/or reduce a noise in the sensor optical radiation signal **138**, thereby improving a signal-to-noise ratio (SNR) of the sensor optical radiation signal **138**. Thus, a performance of the optical biosensor may be improved.

(19) In some embodiments, the optical signal enhancement structure **126** comprises an incident radiation filter layer **128** disposed on a bottom surface of the dielectric structure **116** between the dielectric structure **116** and the glue layer **114**. The incident radiation filter layer **128** is disposed and directly over the micro-lens **112** and directly between the micro-lens and the receptor layer **118**. In some embodiments, the incident radiation filter layer **128** is configured to block the incident optical radiation **136** from impinging on the photodetector **104**. For example, in some embodiments, the incident radiation filter layer **128** is configured to filter (e.g., attenuate) the incident optical radiation **136** and/or some noise radiation (e.g., ambient optical radiation, some other optical background noise, or the like). Thus, a noise (e.g., the incident optical radiation **136**, ambient optical radiation, some other optical background noise, or the like) detected by the photodetector **104** may be reduced and hence a performance of the optical biosensor integrated chip may be improved.

(20) In some embodiments, the semiconductor layer **102** may, for example, comprise silicon or some other suitable material. In some embodiments, the photodetector **104** may, for example, be or comprise a photodiode, a complementary metal-oxide-semiconductor (CMOS) image sensor, an avalanche photodiode (APD), a single photon avalanche diode (SPAD) a charge coupled device (CCD), an ion-sensitive field-effect transistor, an infrared photodetector, or the like. In some embodiments, the glue layer **114** may, for example, comprise a material having a high optical transmission rate. In some embodiments, the glue layer **114** may, for example, comprise an epoxy or some other suitable material. In some embodiments, the incident radiation filter layer **128** may,

for example, comprise a stack of various films having different refractive indices or some other suitable material(s). In some embodiments, the incident radiation filter layer **128** may be or comprise an infrared (IR) radiation filter layer, a near infrared (NIR) radiation filter layer, or the like. In some embodiments, the dielectric structure **116** comprises a dielectric that is substantially optically transparent to the optical radiation emitted by the bioreaction during the operation of the optical biosensor (e.g., the sensor optical radiation signal **138**). In some embodiments, the dielectric structure **116** may, for example, comprise glass, quartz, some plastic material, or some other suitable material. In some embodiments, the receptor layer **118** may, for example, comprise a self-assembled monolayer (SAM), a hydrogel layer, a hydrophilic layer, or some other suitable material. In some embodiments, the capping layer **122** may, for example, comprise glass, quartz, polydimethylsiloxane (PDMS), polymethyl methacrylate (PMMA), some plastic material, or some other suitable material. In some embodiments, the capping layer **122** may be or comprise an electrowetting-on-dielectric (EWOD) microfluidic device, a dielectrophoresis microfluidic device, or some other suitable microfluidic device. In some embodiments, the capping layer **122** may comprise a microfluidic pump (e.g., for pumping the sample solution **130** into the microfluidic channel **124**), a valve (e.g., for controlling a flow of the sample solution **130**), a microfluidic mixer, or some other suitable microfluidic device components.

(21) FIG. **2** illustrates a cross-sectional view **200** of some embodiments of the optical biosensor integrated chip of FIG. **1** in which the optical signal enhancement structure **126** additionally or alternatively comprises an aperture layer **202**.

(22) In some embodiments, the aperture layer **202** is disposed along a bottom surface of the dielectric structure **116** and a top surface of the glue layer **114**. For example, in some embodiments, the aperture layer **202** is disposed on a bottom surface of the incident radiation filter layer **128** and directly between portions of the glue layer **114** and portions of the incident radiation filter layer **128**. One or more sidewalls of the aperture layer **202** delimit an aperture **204** in the aperture layer **202**. For example, in some embodiments (e.g., where the aperture **204** is circular-shaped when viewed from above), the aperture **204** is delimited by a curved sidewall of aperture layer **202** that extends in a circular shape. In some embodiments (e.g., where the aperture **204** is square-shaped or has some other shape when viewed from above), the aperture **204** is delimited by a plurality of sidewalls of the aperture layer **202** (e.g., four sidewalls in embodiments where the aperture **204** is square shaped). In some embodiments (e.g., when the aperture layer **202** is viewed in cross-section), the aperture **204** is delimited by a pair of sidewalls of the aperture layer **202**.

(23) In some embodiments, a portion of the glue layer **114** extends directly between the one or more sidewalls of the aperture layer **202** that delimit the aperture **204**. The portion of the glue layer **114** fills the aperture **204**. The portion of the glue layer is directly over the micro-lens **112** and directly between the micro-lens **112** and the receptor layer **118**. In some embodiments, the portion of the glue layer **114** is directly between the micro-lens **112** and the bioreceptor **120**.

(24) In some embodiments, the aperture layer **202** is configured to limit the optical radiation that can pass through to the photodetector **104**. For example, some optical radiation may be able to pass through the aperture **204** in the aperture layer **202** but not through the aperture layer **202** itself. In some embodiments, the aperture layer **202** can reduce a crosstalk between the photodetector **104** and neighboring photodetectors (not shown). For example, the aperture layer **202** can block optical radiation from a periphery of the photodetector **104** where the photodetector **104** borders the neighboring photodetectors, thereby reducing a likelihood of crosstalk occurring between the photodetector **104** and the neighboring photodetectors. Thus, a performance of the optical biosensor may be improved. Further, in some embodiments, the aperture layer **202** can block some undesired optical radiation (e.g., the incident optical radiation **136**, some ambient optical radiation, some other optical background noise, or the like) from reaching the photodetector **104** which may reduce a noise in the radiation signal detected by the photodetector **104** (e.g., the sensor optical radiation signal **138**) and hence improve a SNR of the detected radiation signal.

(25) In some embodiments, the aperture layer **202** may, for example, comprise aluminum, titanium, titanium nitride, or some other suitable material. In some embodiments, a thickness of the aperture layer **202** ranges from about 100 angstroms to 5000 angstroms or some other suitable range.
(26) FIG. **3** illustrates a cross-sectional view **300** of some embodiments of the optical biosensor integrated chip of FIG. **2** in which the optical signal enhancement structure **126** additionally or alternatively comprises a focusing layer **302**.
(27) The focusing layer **302** is disposed within the dielectric structure **116** directly over the micro-lens **112**. For example, in some embodiments, the dielectric structure **116** comprises a first dielectric layer **304** and a second dielectric layer **306** over the first dielectric layer **304**. Further, the focusing layer **302** is disposed over the first dielectric layer **304** and the second dielectric layer **306** is disposed over the focusing layer **302**. In some embodiments, the focusing layer **302** comprises a material having a suitable refractive index (n) and a suitable extinction coefficient (k). In some embodiments, the focusing layer **302** may, for example, comprise amorphous silicon, silicon nitride, or some other suitable material.
(28) In some embodiments, the focusing layer **302** has a plurality of ring-shaped protrusions (e.g., as shown in the top view **800***a* illustrated in FIG. **8**A and as shown by dashed line **302***a* of FIG. **3**) along a top surface of the focusing layer **302**. For example, in some embodiments, the protrusions of focusing layer **302** are similar to those of a Fresnel lens, a grating lens, or the like. The focusing layer **302** is configured to focus the sensor optical radiation signal (e.g., **138** of FIG. **1**) to enhance the sensor optical radiation signal. For example, in some embodiments, the focusing layer **302** is configured to increase the intensity of the sensor optical radiator signal before the sensor optical radiation signal reaches, and is detected by, the photodetector **104**. Thus, the focusing layer **302** may improve the SNR of the sensor optical radiation signal and hence the performance of the optical biosensors may be improved. In some embodiments, the shape and pitch of the ring-shaped protrusions (e.g., illustrated by dashed line **302***a*) may be adjusted to tune the focus of the focusing layer **302**.
(29) In some other embodiments, sidewalls of the focusing layer **302** are configured to reflect radiation (e.g., the sensor optical radiation signal **138** of FIG. **1**) toward the photodetector **104** to increase an intensity of the radiation. For example, the focusing layer **302** may be or function as a light pipe or a light channel. Thus, the focusing layer **302** may improve the SNR of the signal received by the photodetector **104** (e.g., the sensor optical radiation signal **138** of FIG. **1**) and hence may improve a performance of the photodetector **104**. Further, in some embodiments, the sidewalls of the focusing layer **302** may reflect radiation toward the photodetector **104** that is directly under the focusing layer **302** and away from neighboring photodetectors (not shown) that are not directly under the focusing layer **302** to reduce a crosstalk between the underlying photodetector **104** and the neighboring photodetectors. In some embodiments, the sidewalls of the focusing layer **302** are slanted (e.g., as shown by dashed lines **302***b*) such that a distance between the sidewalls along the tops of the sidewalls is greater than a distance between the sidewalls along the bottoms of the sidewalls. The slanting of the sidewalls may be controlled to tune the angle of reflection of the radiation to further increase the intensity of the radiation that is reflected toward the photodetector **104**.
(30) FIG. **4** illustrates a cross-sectional view **400** of some embodiments of the optical biosensor integrated chip of FIG. **3** in which the optical signal enhancement structure **126** additionally or alternatively comprises a reflector layer **402**.
(31) The reflector layer **402** is disposed along a top surface of the dielectric structure **116**. One or more sidewalls of the reflector layer **402** delimit an aperture **406** in the reflector layer **402**. In some embodiments (e.g., when the reflector layer **402** is viewed in cross-section), the aperture **406** is delimited by a pair of sidewalls of the reflector layer **402**. A first clad layer **404** is disposed over the reflector layer **402**. In some embodiments, a portion of the first clad layer **404** extends directly between the sidewalls of the reflector layer **402** that delimit the aperture **406**. The portion of the

first clad layer **404** fills the aperture **406**. The portion of the first clad layer **404** is directly over the micro-lens **112** and directly between the micro-lens **112** and the receptor layer **118**. In some embodiments, the portion of the first clad layer **404** is on the top surface of the dielectric structure **116**. In some embodiments, the portion of the first clad layer **404** is directly between the micro-lens **112** and the bioreceptor **120**.
(32) In some embodiments, the reflector layer **402** is configured to reflect noise radiation (e.g., the incident optical radiation **136**, some ambient optical radiation, some other optical background noise, or the like) while allowing the sensor optical radiation signal **138** to pass through at the apertures **406**. As a result, an SNR of the sensor optical radiation signal **138** may be improved and hence a performance of the optical biosensor may be improved. In some embodiments, by including the reflector layer **402** over the focusing layer **302** such that the aperture **406** is directly over the focusing layer **302**, the noise radiation may be blocked from the focusing layer **302** without blocking the sensor optical radiation signal **138**, thereby reducing the likelihood that the noise radiation will be intensified and focused on the photodetector **104** by the focusing layer **302**.
(33) In some embodiments, the reflector layer **402** may, for example, comprise aluminum, titanium, titanium nitride, chromium, or some other suitable material. In some embodiments, the reflector layer **402** comprises a different material than the aperture layer **202**. In some embodiments, a thickness of the reflector layer **402** ranges from about 100 angstroms to 2000 angstroms or some other suitable range. In some embodiments, the thickness of the reflector layer **402** is less than the thickens of the aperture layer **202**. In some embodiments, the aperture **406** in the reflector layer **402** is smaller than the aperture **204** in the aperture layer **202**. For example, when viewed in cross-section, a distance between the sidewalls of the reflector layer **402** that delimit the aperture **406** is less than a distance between the sidewalls of the aperture layer **202** that delimit the aperture **204**. In some embodiments, the first clad layer **404** may, for example, comprise a low-k dielectric (e.g., silicon dioxide or the like) or some other suitable material.
(34) FIG. **5** illustrates a cross-sectional view **500** of some embodiments of the optical biosensor integrated chip of FIG. **4** in which a waveguide channel core layer **502** is disposed between the optical signal enhancement structure **126** and the receptor layer **118**.
(35) The waveguide channel core layer **502** is disposed over the first clad layer **404** and a second clad layer **504** is disposed over the waveguide channel core layer **502**. Sidewalls of the second clad layer **504** and an upper surface of the waveguide channel core layer **502** delimit a bioreaction chamber **506**. The bioreaction chamber **506** is directly over the photodetector **104**. In some embodiments, the receptor layer **118** lines the sidewalls of the second clad layer **504** and the upper surface of the waveguide channel core layer **502** and the receptor layer **118** further delimits the bioreaction chamber **506**. In some embodiments, the bioreceptor **120** is disposed along the receptor layer **118** directly between the sidewalls of the second clad layer **504** and directly over the upper surface of the waveguide channel core layer **502**.
(36) In some embodiments, the incident optical radiation **136** travels along the waveguide channel core layer **502** to the bioreaction chamber **506** by way of total internal reflection. The incident optical radiation impinges on the bioreceptor **120** at the bioreaction chamber **506** to excite a bioreaction at the bioreceptor **120**. The bioreaction may then emit the sensor optical radiation signal **138** which the photodetector **104** is configured to detect.
(37) In some embodiments, a temperature sensor **508** and/or a heater **510** are disposed within the dielectric structure **116**. For example, in some embodiments, the dielectric structure **116** includes the first dielectric layer **304**, the second dielectric layer **306** over the first dielectric layer **304**, and a third dielectric layer **512** over the second dielectric layer **306**. Further, the temperature sensor **508** and the heater **510** are disposed over the second dielectric layer **306** and the third dielectric layer **512** is disposed over both the temperature sensor **508** and the heater **510**. The temperature sensor **508** and the heater **510** are respectively configured to detect and control the temperature of the integrated chip and/or the sample solution (e.g., **130** of FIG. **1**) injected into the microfluidic

channel **124**. By detecting and controlling the temperature of the integrated chip and/or the sample solution that is injected into the integrated chip, a performance of the optical biosensor integrated chip may be improved.
(38) In some embodiments, a trench isolation structure **514** is disposed on opposite sides of the photodetector **104** and isolates the photodetector **104** from neighboring photodetectors (not shown). In some embodiments, the trench isolation structure **514** may, for example, comprise a dielectric, a metal, or some other suitable material. In some embodiments, a composite metal grid (CMG) structure **516** is disposed on opposite sides of the color filter **106** and separates the color filter **106** from neighboring color filters (not shown). In some embodiments, the CMG structure **516** is disposed directly over the trench isolation structure **514**. In some embodiments, the CMG structure **516** may, for example, comprises a metal and a dielectric or some other suitable material(s).
(39) In some embodiments, the waveguide channel core layer **502** may, for example, comprise a high-k dielectric (e.g., barium strontium titanate (BST), lead zirconate titanate (PZT), tantalum oxide, hafnium oxide, silicon nitride, aluminum oxide) or some other suitable material. In some embodiments, the second clad layer **504** may, for example, comprise a same material as the first clad layer **404**. In some other embodiments, the second clad layer **504** may comprise a different low-k dielectric than the first clad layer **404**. In some embodiments, the temperature sensor **508** and/or the heater **510** may, for example, comprise platinum, polysilicon, or some other suitable material.
(40) FIG. **6** illustrates a cross-sectional view **600** of some embodiments of an optical biosensor integrated chip including a receptor layer **118** disposed directly over a plurality of photodetectors **104** and an optical signal enhancement structure **126** disposed between the receptor layer **118** and the plurality of photodetectors **104**.
(41) A plurality of color filters **106** are disposed over the plurality of photodetectors **104**, respectively, and a plurality of micro-lenses **112** are disposed over the plurality of color filters **106**, respectively. In some embodiments, the aperture layer **202** comprises a plurality of aperture segments (not labeled) when viewed in cross-section. Sidewalls of the aperture layer **202** delimit a plurality of apertures **204** in the aperture layer **202**. In some embodiments, the focusing layer **302** comprises a plurality of focusing layer segments (not labeled). In some embodiments, each of the focusing layer segments has a plurality of ring-shaped protrusions along the top surfaces of the segments. In some other embodiments, the focusing layer segments have slanted sidewalls. In some embodiments, a plurality of temperature sensors **508** and a plurality of heaters **510** are disposed over the second dielectric layer **306** and are laterally spaced apart from one another. In some embodiments, the reflector layer **402** comprises a plurality of reflector segments (not labeled) when viewed in cross-section. Sidewalls of the reflector layer **402** delimit a plurality of apertures **406** in the reflector layer **402**.
(42) In some embodiments, the waveguide channel core layer **502** comprises a light coupler structure **602**. The light coupler structure **602** is formed by a plurality of segments of the waveguide channel core layer **502**. In some embodiments, the waveguide channel core layer **502** further comprises a light splitter structure (not shown) configured to transfer (e.g., split) optical radiation into multiple output waveguide channel core layers from an input waveguide channel core layer.
(43) A plurality of bioreaction chambers **506** are disposed along the second clad layer **504** and the waveguide channel core layer **502**. The bioreaction chambers **506** are directly over the apertures **406** in the reflector layer **402**, the focusing segments of the focusing layer **302**, the incident radiation filter layer **128**, and the apertures **204** in the aperture layer **202**. Further, the bioreaction chambers **506** are directly over one or more of the micro-lenses **112**, one or more of the color filters **106**, and one or more of the photodetectors **104**. In some embodiments, a plurality of microbeads **604** having bioreceptors **120** disposed thereon are disposed along the receptor layer **118** in the plurality of bioreaction chambers **506**, respectively. In some other embodiments, the biosensor is

devoid of the microbeads **604** and the bioreceptors **120** are disposed on the receptor layer **118**.
(44) In some embodiments, a carrier wafer **606** is disposed below the ILD structure **108**. An isolation layer **608** is disposed along a bottom surface and sidewalls of the carrier wafer **606**. A through-substrate via (TSV) layer **610** comprising a plurality of separate TSV segments (not labeled) is disposed along sidewalls and a bottom surface of the isolation layer **608**. The TSV segments are coupled to conductive features of the interconnect structure **110**. The isolation layer **608** isolates the TSV layer **610** from the carrier wafer **606**. A passivation layer **612** is disposed on sidewalls and a bottom surface of the TSV layer **610**. The passivation layer **612** extends between and isolates TSV segments of the TSV layer **610**. Openings in the passivation layer **612** expose portions of the TSV segments of the TSV layer **610**.
(45) In some embodiments, bioreactions between the bioreceptors **120** and the analytes **132** are excited by first incident radiation **136***a* that is emitted directly over the bioreaction chambers **506** and that passes through the capping layer **122** to the bioreaction chambers **506** to excite bioreactions at each of the bioreaction chambers **506**. In some other embodiments, bioreactions between the bioreceptors **120** and the analytes **132** are excited by second incident radiation **136***b* which passes through the capping layer **122** and the second clad layer **504** and impinges on the light coupler structure **602**. The second incident radiation **136***b* then travels along the waveguide channel core layer **502** from the light coupler structure **602** to each of the bioreaction chambers **506**. For example, after impinging on the light coupler structure **602**, the second incident radiation **136***b* experiences total internal reflection within the waveguide channel core layer **502** such that photons from the second incident radiation **136***b* travel along the waveguide channel core layer **502** and impinge on the bioreceptors **120** and analytes **132** at each of the bioreaction chambers **506**. The second incident radiation **136***b* excites bioreactions at each of the bioreaction chambers **506**.
(46) In some embodiments, the microbeads **604** may, for example, comprise silicon, silica, polystyrene, copper, a magnetic material, or some other suitable material. In some embodiments, the carrier wafer **606** may, for example, comprise silicon or some other suitable material. In some embodiments, the isolation layer **608** may, for example, comprise silicon dioxide, silicon nitride, or some other suitable material. In some embodiments, the TSV layer **610** may, for example, comprise copper, tungsten, aluminum, or some other suitable material. In some embodiments, the passivation layer **612** may, for example, comprise silicon nitride or some other suitable material.
(47) FIGS. **7**, **8**A, **8**B, **9**, and **10** illustrate top views **700**, **800***a*, **800***b*, **900**, **1000** of some embodiments of the optical biosensor integrated chip of FIG. **6**.
(48) In some embodiments, top view **700** illustrated in FIG. **7** is taken across line A-A′ of FIG. **6**. In some embodiments, the apertures **204** in the aperture layer **202** are circular-shaped when viewed from above, as illustrated in FIG. **7**. In some other embodiments (not shown), the apertures **204** in the aperture layer **202** are square-shaped or have some other shape when viewed from above.
(49) In some embodiments, top views **800***a*, **800***b* illustrated in FIGS. **8**A and **8**B are taken across line B-B′ of FIG. **6**. In some embodiments (e.g., as illustrated in top view **800***a* of FIG. **8**A), the focusing segments of the focusing layer **302** have a plurality of ring-shaped protrusions along the top surfaces of the segments. In some embodiments, the focusing segments of the focusing layer **302** are square-shaped, circular-shaped, or have some other shape when viewed from above. In some other embodiments (e.g., as illustrated in top view **800***b* of FIG. **8**B), the focusing segments of the focusing layer **302** have substantially planar top surfaces and are square-shaped, rectangular-shaped, or have some other shape.
(50) In some embodiments, top view **900** illustrated in FIG. **9** is taken across line C-C′ of FIG. **6**. In some embodiments, the apertures **406** in the reflector layer **402** are circular-shaped when viewed from above, as illustrated in FIG. **9**. In some other embodiments (not shown), the apertures **406** in the reflector layer **402** are square-shaped or have some other shape when viewed from above.
(51) In some embodiments, top view **1000** illustrated in FIG. **10** is taken across line D-D′ of FIG. **6**. The light coupler structure **602** is formed by a plurality of adjacent segments of the waveguide

channel core layer **502**. In some embodiments, the segments of the waveguide channel core layer **502** that form the light coupler structure **602** have varying lengths.

(52) FIGS. **11-39** illustrate cross-sectional views **1100-3900** of some embodiments of a method for forming an optical biosensor integrated chip including a receptor layer **118** disposed directly over a photodetector **104** and an optical signal enhancement structure **126** disposed between the receptor layer **118** and the photodetector **104**. Although FIGS. **1100-3900** are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **1100-3900** are not limited to such a method, but instead may stand alone as structures independent of the method.

(53) FIGS. **11-16** illustrate cross-sectional views **1100-1600** of some embodiments of a method for forming an image sensor wafer.

(54) As shown in cross-sectional view **1100** of FIG. **11**, a plurality of photodetectors **104** are formed within a semiconductor layer **102** along a frontside **102***a* of the semiconductor layer **102**. In some embodiments, the semiconductor layer **102** is formed on a substrate **1102** before the photodetectors **104** are formed within the semiconductor layer **102**.

(55) In some embodiments, forming the plurality of photodetectors **104** comprises performing one or more implantation processes, diffusion processes, or some other suitable processes along the semiconductor layer **102**. In some embodiments, the semiconductor layer **102** comprises silicon or some other suitable material and is formed on the substrate **1102** by a chemical vapor deposition (CVD) process, a physical vapor deposition (PVD) process, an atomic layer deposition (ALD) process, an epitaxial growth process, or some other suitable process. In some embodiments, the substrate **1102** comprises silicon, a semiconductor-on-insulator (SOI) stack, or some other suitable material(s).

(56) As shown in cross-sectional view **1200** of FIG. **12**, an ILD structure **108** is formed over a backside **102***b* of the semiconductor layer **102**, opposite the frontside **102***a*. Further, an interconnect structure **110** is formed within the ILD structure **108**. The interconnect structure **110** comprises a plurality of conductive features disposed within the ILD structure **108**.

(57) In some embodiments, the ILD structure **108** comprises silicon oxide, silicon nitride, or some other suitable material and is formed by a CVD process, a PVD process, an ALD process, or some other suitable process. In some embodiments, the conductive features of the interconnect structure **110** comprise copper, tungsten, aluminum, or some other suitable material and are formed within the ILD structure **108** by a CVD process, a PVD process, an ALD process, a sputtering process, an electroless deposition (ELD) process, an electrochemical deposition (ECD) process, or some other suitable process.

(58) As shown in cross-sectional view **1300** of FIG. **13**, a carrier wafer **606** is bonded over the ILD structure **108**. For example, in some embodiments, the carrier wafer **606** is bonded to the ILD structure **108** along a first side of the carrier wafer **606**. In some other embodiments, the carrier wafer **606** is bonded to a bonding layer (not shown) that is over the ILD structure **108** along the first side of the carrier wafer **606**. In some embodiments, the carrier wafer **606** comprises silicon or some other suitable material. In some embodiments, the bonding comprises a fusion boding process, an adhesive bonding process, or some other suitable process.

(59) As shown in cross-sectional view **1400** of FIG. **14**, the substrate **1102** is removed from the backside **102***b* of the semiconductor layer **102**. In some embodiments, the semiconductor layer **102** is thinned along the backside **102***b* of the semiconductor layer **102**. In some embodiments, the removal of the substrate **1102** from the backside **102***b* of the semiconductor layer **102** and/or the thinning of the semiconductor layer **102** comprises a chemical mechanical planarization (CMP) process, a grinding process, a blanket etch back process, or some other suitable process.

(60) As shown in cross-sectional view **1500** of FIG. **15**, a plurality of color filters **106** are formed over the plurality of photodetectors **104**, respectively, along the backside **102***b* of the semiconductor layer **102**. In some embodiments, forming the color filters **106** comprises depositing and patterning a plurality of color resist layers over the photodetectors **104**. Further, a plurality of

micro-lenses **112** are formed directly over the plurality of color filters **106**, respectively. In some embodiments, forming the micro-lenses **112** comprises disposing a micro-lens template over a micro-lens material and shaping the micro-lens material according to the micro-lens template.

(61) As shown in cross-sectional view **1600** of FIG. **16**, a glue layer **114** is deposited over the micro-lenses **112**. Further, the glue layer **114** is deposited on opposite sides of the micro-lenses **112** and color filters **106**. In some embodiments, the glue layer **114** comprises an epoxy or some other suitable material and is deposited by a spread coating process, a CVD process, a PVD process, an ALD process, or some other suitable process. In some other embodiments, the glue layer **114** may be deposited around a periphery of the color filters **106** and micro-lenses **112** but not directly over the micro-lenses **112**.

(62) FIGS. **17-34** illustrate cross-sectional views **1700-3400** of some embodiments of a method for forming a bioreaction wafer.

(63) As shown in cross-sectional view **1700** of FIG. **17**, an incident radiation filter layer **128** is deposited over a first side of a first dielectric layer **304**. In some embodiments, the first dielectric layer **304** comprises glass, quartz, some plastic material, or some other suitable material. In some embodiments, the incident radiation filter layer **128** comprises a stack of various films having different refractive indices or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(64) As shown in cross-sectional view **1800** of FIG. **18**, an aperture layer **202** is deposited over the incident radiation filter layer **128**. In some embodiments, the aperture layer **202** comprises aluminum, titanium, titanium nitride, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, a sputtering process, an ELD process, an ECD process, or some other suitable process.

(65) As shown in cross-sectional view **1900** of FIG. **19**, the aperture layer **202** is patterned to form a plurality of apertures **204** in the aperture layer **202**. The plurality of apertures **204** are delimited by sidewalls of the aperture layer **202**. In some embodiments, the patterning comprises forming a masking layer (not shown) over the aperture layer **202** and etching the aperture layer **202** according to the masking layer to form the apertures **204** in the aperture layer **202**. In some embodiments, the etching comprises a dry etching process (e.g., a plasma etching process, a reaction ion etching process, an ion beam etching process, or the like) or some other suitable process. In some embodiments, the masking layer may be removed during and/or after the etching.

(66) As shown in cross-sectional view **2000** of FIG. **20**, a focusing layer **302** is deposited over a second side of the first dielectric layer **304**, opposite the first side. In some embodiments, the focusing layer **302** comprises amorphous silicon, silicon nitride, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(67) As shown in cross-sectional view **2100** of FIG. **21**, the focusing layer **302** is patterned to form a plurality of focusing layer segments (not labeled). In some embodiments, the patterning comprises performing one or more etching processes according to one or more masking layers. In some embodiments, the patterning forms a plurality of protrusions along top surfaces of the focusing layer segments (e.g., as illustrated in FIG. **8**A and by dashed line **302***a* of FIG. **3**). In some other embodiments, the patterning may result in the focusing layer segments having slanted sidewalls (e.g., as illustrated by dashed lines **302***b* of FIG. **3**) and/or substantially planar top surfaces (e.g., as illustrated in FIG. **8**B).

(68) As shown in cross-sectional view **2200** of FIG. **22**, a second dielectric layer **306** is deposited over the focusing layer **302** and between the focusing layer segments (not labeled). In some embodiments, the second dielectric layer **306** comprises glass, quartz, some plastic material, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(69) As shown in cross-sectional view **2300** of FIG. **23**, a thermal controller layer **2302** is deposited

over the second dielectric layer **306**. In some embodiments, the thermal controller layer **2302** comprises platinum, polysilicon, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, a sputtering process, an ELD process, an ECD process, or some other suitable process.

(70) As shown in cross-sectional view **2400** of FIG. **24**, the thermal controller layer (e.g., **2302** of FIG. **23**) is patterned to form a plurality of temperature sensors **508** and a plurality of heaters **510** from the thermal controller layer. In some embodiments, the patterning comprises forming a masking layer (not shown) over the thermal controller layer and etching the thermal controller layer according to the masking layer to form the temperature sensors **508** and the heaters **510** from the thermal controller layer. In some embodiments, the etching comprises a dry etching process or some other suitable process. In some embodiments, the masking layer may be removed during and/or after the etching.

(71) As shown in cross-sectional view **2500** of FIG. **25**, a third dielectric layer **512** is deposited over the temperature sensors **508**, over the heaters **510**, and between the temperature sensors **508** and the heaters **510**. In some embodiments, the third dielectric layer **512** comprises glass, quartz, some plastic material, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(72) As shown in cross-sectional view **2600** of FIG. **26**, a reflector layer **402** is deposited over the third dielectric layer **512**. In some embodiments, the reflector layer **402** comprises aluminum, titanium, titanium nitride, chromium, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, a sputtering process, an ELD process, an ECD process, or some other suitable process.

(73) As shown in cross-sectional view **2700** of FIG. **27**, the reflector layer **402** is patterned to form a plurality of apertures **406** in the reflector layer **402**. The plurality of apertures **406** are delimited by sidewalls of the reflector layer **402**. In some embodiments, the patterning comprises forming a masking layer (not shown) over the reflector layer **402** and etching the reflector layer **402** according to the masking layer to form the apertures **406** in the reflector layer **402**. In some embodiments, the etching comprises a dry etching process or some other suitable process. In some embodiments, the masking layer may be removed during and/or after the etching.

(74) As shown in cross-sectional view **2800** of FIG. **28**, a first clad layer **404** is deposited over the reflector layer **402** and between the sidewalls of the reflector layer **402** that delimit the apertures **406** in the reflector layer **402**. In some embodiments, the first clad layer **404** comprises silicon dioxide or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(75) As shown in cross-sectional view **2900** of FIG. **29**, a waveguide channel core layer **502** is deposited over the first clad layer **404**. In some embodiments, the waveguide channel core layer **502** comprises BST, PZT, tantalum oxide, hafnium oxide, silicon nitride, aluminum oxide, or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process.

(76) As shown in cross-sectional view **3000** of FIG. **30**, the waveguide channel core layer **502** is patterned to form a light coupler structure **602** along the waveguide channel core layer **502**. In some embodiments, the waveguide channel core layer **502** is also patterned to form a light splitter (not shown) along the waveguide channel core layer **502**. In some embodiments, the patterning comprises forming a masking layer (not shown) over the waveguide channel core layer **502** and etching the waveguide channel core layer **502** according to the masking layer to form the light coupler structure **602** and/or the light splitter (not shown) along the waveguide channel core layer **502**. In some embodiments, the etching comprises a dry etching process or some other suitable process. In some embodiments, the masking layer may be removed during and/or after the etching.

(77) As shown in cross-sectional view **3100** of FIG. **31**, a second clad layer **504** is deposited over the waveguide channel core layer **502** and between portions of the waveguide channel core layer

502 at the light coupler structure **602**. In some embodiments, the second clad layer **504** comprises silicon dioxide or some other suitable material and is deposited by a CVD process, a PVD process, an ALD process, or some other suitable process. In some embodiments, a top surface of the second clad layer **504** is planarized (e.g., by way of a CMP process or some other suitable process) after the second clad layer **504** is deposited.

(78) As shown in cross-sectional view **3200** of FIG. **32**, the second clad layer **504** is patterned to form a plurality of bioreaction chambers **506** along the second clad layer **504** and directly over the apertures **406**, the focusing layer segments of the focusing layer **302**, and the apertures **204**. The patterning uncovers portions of an upper surface of the waveguide channel core layer **502**. The bioreaction chambers **506** are delimited by sidewalls of the second clad layer **504** and the portions of the upper surface of the waveguide channel core layer **502** that were uncovered by the patterning. In some embodiments, the patterning comprises forming a masking layer (not shown) over the second clad layer **504** and etching the second clad layer **504** according to the masking layer to form the bioreaction chambers **506** along the second clad layer **504**. In some embodiments, the etching comprises a dry etching process or some other suitable process. In some embodiments, the masking layer may be removed during and/or after the etching.

(79) As shown in cross-sectional view **3300** of FIG. **33**, a receptor layer **118** is formed on the second clad layer **504** and on the waveguide channel core layer **502** at the bioreaction chambers **506**. For example, the receptor layer **118** is formed on the sidewalls of the second clad layer **504** and the portions of the upper surface of the waveguide channel core layer **502** that delimit the bioreaction chambers **506**. In some embodiments, the receptor layer **118** is further formed on a top surface of the second clad layer **504**.

(80) In some embodiments, the receptor layer **118** comprises a hydrogel, a hydrophilic coating, or some other suitable material and is deposited on the second clad layer **504** and on the waveguide channel core layer **502** at the bioreaction chambers **506**. In some other embodiments, the receptor layer **118** comprises a self-assembled monolayer (SAM) or the like which is formed on the second clad layer **504** and on the waveguide channel core layer **502** at the bioreaction chambers **506** by a SAM formation process.

(81) In some embodiments, bioreceptors **120** are subsequently immobilized on the receptor layer **118** at the bioreaction chambers **506**. In some other embodiments, the bioreceptors **120** may be immobilized at the bioreaction chambers **506** when a sample solution (e.g., **130** of FIG. **39**) is provided to the bioreaction chambers **506** (see, for example, FIG. **39**).

(82) As shown in cross-sectional view **3400** of FIG. **34**, a capping layer **122** is bonded over the second clad layer **504** to form a microfluidic channel **124** over the bioreaction chambers **506**. In some embodiments, the capping layer **122** is bonded to the second clad layer **504**. In some embodiments (not shown), the second clad layer **504** is bonded to an adhesive bonding layer (not shown) that is disposed on a top surface the second clad layer **504**. In some embodiments (not shown), the receptor layer **118** is disposed directly between the capping layer **122** and the second clad layer **504** along the bonding interface.

(83) In some embodiments, the bonding comprises a fusion bonding process, a plasma activated bonding process, a surface activated bonding process, an anodic bonding process, an adhesive bonding process, or some other suitable process. In some embodiments, the capping layer **122** comprises glass, quartz, polydimethylsiloxane (PDMS), polymethyl methacrylate (PMMA), some plastic material, or some other suitable material. In some embodiments, before the bonding, the capping layer **122** is patterned to form a capping recess (not labeled), the inlet **122***a*, and the outlet **122***b* in the capping layer.

(84) As shown in cross-sectional view **3500** of FIG. **35**, the biosensor wafer (not labeled) is bonded over the image sensor wafer (not labeled). For example, the biosensor wafer is bonded to the image sensor wafer along the glue layer **114**, the aperture layer **202**, and the incident radiation filter layer **128**. In some embodiments, the glue layer **114** comprises an epoxy configured to adhere to the

aperture layer **202** and the incident radiation filter layer **128**. In some embodiments, the bonding comprises an adhesive bonding process, a fusion bonding process, or some other suitable process. In some embodiments (e.g., in which the glue layer **114** is disposed around a periphery of the color filters **106** and the micro-lenses **112** but not directly over the micro-lenses **112**), the glue layer **114**, the aperture layer **202**, and the incident radiation filter layer **128** are bonded along the periphery of the color filters **106** and the micro-lenses **112**. In some embodiments, a cavity (not shown) comprising air or the like exists directly over the micro-lenses **112** and directly between the micro-lenses **112** and the aperture **202**.

(85) As shown in cross-sectional view **3600** of FIG. **36**, the carrier wafer **606** is thinned along a second side of the carrier wafer **606**, opposite the first side. In some embodiments, the thinning comprise a grinding process, an etch back process, a CMP process, or some other suitable process.

(86) As shown in cross-sectional view **3700** of FIG. **37**, a through-substrate via (TSV) structure comprising a TSV layer **610** is formed over the second side of the carrier wafer **606** such that the TSV layer **610** is coupled to conductive features of the interconnect structure **110**. In some embodiments, forming the TSV structure comprises etching the carrier wafer **606** and the ILD structure **108** to form TSV openings (not shown) in the carrier wafer **606** and the ILD structure **108**. The TSV openings uncover conductive features of the interconnect structure **110**. An isolation layer **608** is deposited over the second side of the carrier wafer **606** and along the TSV openings. The isolation layer **608** is etched to remove the isolation layer **608** from the conductive features of the interconnect structure **110**. The TSV layer **610** is deposited over the second side of the carrier wafer **606**, along the TSV openings, and on the conductive features of the interconnect structure **110**. The TSV layer **610** is etched to separate the TSV layer **610** in separate TSV segments. A passivation layer **612** is deposited over the TSV layer **610** and in remainders of the TSV openings to fill the TSV openings. The passivation layer **612** is etched to form passivation openings (not labeled) in the passivation layer **612**. The passivation openings uncover portions of the TSV segments of the TSV layer **610**.

(87) As shown in cross-sectional view **3800** of FIG. **38**, a dicing process is performed along a scribe line **3802** to dice the bonded wafers into individual die. In some embodiments, the dicing process comprises bringing a dicing blade into contact with the wafers along the scribe line **3802** to cut through the wafers.

(88) As shown in cross-sectional view **3900** of FIG. **39**, in some embodiments, a sample solution **130** carrying microbeads **604** with bioreceptors **120** immobilized thereon is injected into the microfluidic channel **124** via the inlet **122***a*. In some embodiments, the sample solution **130** may also carry analytes (e.g., **132** of FIG. **6**). In some other embodiments, a separate sample solution carrying the analytes is subsequently injected into the microfluidic channel **124** via the inlet **122***a*.

(89) FIG. **40** illustrates a flow diagram of some embodiments of a method **4000** for forming an optical biosensor integrated chip. While method **4000** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(90) At block **4002**, form an image sensor wafer. In some embodiments, block **4002** comprises block **4002***a*, block **4002***b*, block **4002***c*, and block **4002***d*. FIGS. **11-16** illustrate cross-sectional views **1100-1600** of some embodiments corresponding to block **4002**.

(91) At block **4002***a*, form a plurality of photodetectors along a semiconductor layer. FIG. **11** illustrates a cross-sectional view **1100** of some embodiments corresponding to block **4002***a*.

(92) At block **4002***b*, form a plurality of color filters over the plurality of photodetectors. FIG. **15** illustrates a cross-sectional view **1500** of some embodiments corresponding to block **4002***b*.

(93) At block **4002***c*, form a plurality of micro-lenses over the plurality of color filters. FIG. **15** illustrates a cross-sectional view **1500** of some embodiments corresponding to block **4002***c*.
(94) At block **4002***d*, deposit a glue layer over the plurality of micro-lenses. FIG. **16** illustrates a cross-sectional view **1600** of some embodiments corresponding to block **4002***d*.
(95) At block **4004**, form a bioreaction wafer. In some embodiments, block **4004** comprises block **4004***a*, block **4004***b*, block **4004***c*, block **4004***d*, and block **4004***e*. FIGS. **17-34** illustrate cross-sectional views **1700-3400** of some embodiments corresponding to block **4004**.
(96) At block **4004***a*, form an optical signal enhancement structure along a dielectric structure. FIGS. **17-27** illustrate cross-sectional views **1700-2700** of some embodiments corresponding to block **4004***a*.
(97) At block **4004***b*, form a waveguide channel structure over the dielectric structure and over the optical signal enhancement structure. In some embodiments, forming the waveguide channel structure comprises depositing a first clad layer over the dielectric structure, depositing a waveguide channel core layer over the first clad layer, patterning the waveguide channel core layer (e.g., to form a light coupler, a light splitter, or the like along the waveguide channel core layer), and depositing a second clad layer over the patterned waveguide channel core layer. FIGS. **28-31** illustrate cross-sectional views **2800-3100** of some embodiments corresponding to block **4004***b*.
(98) At block **4004***c*, form a plurality of bioreaction chambers over the waveguide channel structure. FIG. **32** illustrates a cross-sectional view **3200** of some embodiments corresponding to block **4004***c*.
(99) At block **4004***d*, deposit a receptor layer along the plurality of bioreaction chambers. FIG. **33** illustrates a cross-sectional view **3300** of some embodiments corresponding to block **4004***d*. In some embodiments, method **4000** comprises immobilizing bioreceptors on the receptor layer at the bioreaction chambers during or after the deposition of the receptor layer. In some embodiments, method **4000** alternatively comprises injecting a sample solution carrying microbeads with bioreceptors immobilized thereon into the microfluidic channel after the bioreaction wafer and the image sensor wafer are bonded.
(100) At block **4004***e*, bond a capping layer over the dielectric structure to form a microfluidic channel over the receptor layer and along the plurality of bioreaction chambers. In some embodiments, the microfluidic channel can be formed in the capping layer before the capping layer is bonded over the dielectric structure. FIG. **34** illustrates a cross-sectional view **3400** of some embodiments corresponding to block **4004***e*.
(101) At block **4006**, bond the bioreaction wafer over the image sensor wafer along the glue layer. FIG. **35** illustrates a cross-sectional view **3500** of some embodiments corresponding to block **4006**.
(102) In some embodiments, method **4000** further comprises forming an ILD structure and an interconnect structure within the ILD structure along the semiconductor layer. In some embodiments, method **4000** further comprises forming temperature sensors and/or heaters within the dielectric structure. In some embodiments, method **4000** further comprises forming a TSV layer along the ILD layer and contacting the interconnect structure. In some embodiments, method **4000** further comprises dicing the bonded wafers to separate the bonded wafers into individual die.
(103) Thus, the present disclosure relates to an optical biosensor device comprising a bioreaction device integrated with an image sensor for improving a portability of the optical biosensor device.
(104) Accordingly, in some embodiments, the present disclosure relates to an integrated chip including a semiconductor layer and a photodetector disposed along the semiconductor layer. A color filter is over the photodetector. A micro-lens is over the color filter. A dielectric structure comprising one or more dielectric layers is over the micro-lens. A receptor layer is over the dielectric structure. An optical signal enhancement structure is disposed along the dielectric structure and between the receptor layer and the micro-lens.
(105) In other embodiments, the present disclosure relates to an integrated chip including a semiconductor layer. A plurality of photodetectors are disposed along the semiconductor layer. A

plurality of color filters are over the plurality of photodetectors. A plurality of micro-lenses are over the plurality of color filters. A dielectric structure comprising one or more dielectric layers is over the plurality of micro-lenses. A first clad layer is over the dielectric structure. The first clad layer delimits a plurality of bioreaction chambers that are spaced apart along a top surface of the first clad layer. A receptor layer is over the first clad layer and lines the first clad layer at the plurality of bioreaction chambers. A capping layer is over the first clad layer and over the receptor layer. A microfluidic channel extends through the capping layer along a top surface of the receptor layer and along the plurality of bioreaction chambers. An optical signal enhancement structure is disposed between the plurality of bioreaction chambers and the plurality of photodetectors and is configured to enhance an optical signal emitted from bioreactions occurring along the plurality of bioreaction chambers.

(106) In yet other embodiments, the present disclosure relates to a method for forming an integrated chip. The method includes forming a plurality of photodetectors along a semiconductor layer. A plurality of color filters are formed over the plurality of photodetectors, respectively. A plurality of micro-lenses are formed over the plurality of color filters, respectively. A glue layer is deposited over the plurality of micro-lenses. An optical signal enhancement structure is formed along a dielectric structure. A plurality of bioreaction chambers are formed over the dielectric structure and over the optical signal enhancement structure. A receptor layer is deposited along the plurality of bioreaction chambers. A capping layer having a microfluidic channel therein is bonded over the receptor layer and over the plurality of bioreaction chambers. The dielectric structure is bonded over the plurality of photodetectors and along the glue layer such that the plurality of bioreaction chambers are disposed over the plurality of photodetectors.

(107) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming an integrated chip, the method comprising: forming a plurality of photodetectors along a semiconductor layer; forming a plurality of color filters over the plurality of photodetectors, respectively; forming a plurality of micro-lenses over the plurality of color filters, respectively; depositing a glue layer over the plurality of micro-lenses; forming an optical signal enhancement structure along a dielectric structure; forming a plurality of bioreaction chambers over the dielectric structure and over the optical signal enhancement structure; depositing a receptor layer along the plurality of bioreaction chambers; bonding a capping layer having a microfluidic channel therein over the receptor layer and over the plurality of bioreaction chambers; and bonding the dielectric structure over the plurality of photodetectors and along the glue layer such that the plurality of bioreaction chambers are disposed over the plurality of photodetectors.

2. The method of claim 1, wherein forming the optical signal enhancement structure comprises: depositing an incident radiation filter layer on a backside of the dielectric structure; depositing an aperture layer on the incident radiation filter layer; patterning the aperture layer to form a plurality of first apertures in the aperture layer; forming a focusing layer comprising a plurality of focusing segments within the dielectric structure; depositing a reflector layer over the dielectric structure; and patterning the reflector layer to form a plurality of second apertures in the reflector layer.

3. The method of claim 2, further comprising: forming a temperature sensor and a heater within the

dielectric structure.
4. The method of claim 1, further comprising: depositing a first clad layer over the dielectric structure; and depositing a waveguide channel core layer over the first clad layer; patterning the waveguide channel core layer to form a light coupler structure along the waveguide channel core layer; and depositing a second clad layer over the waveguide channel core layer.
5. The method of claim 4, wherein the plurality of bioreaction chambers are formed by patterning the second clad layer, and wherein the plurality of bioreaction chambers are delimited by sidewalls of the second clad layer and an upper surface of the waveguide channel core layer.
6. A method of forming an integrated chip, the method comprising: forming a plurality of photodetectors along a semiconductor layer; forming a plurality of color filters over the plurality of photodetectors; forming a plurality of micro-lenses over the plurality of color filters; forming an optical signal enhancement structure along a dielectric structure; depositing a first clad layer over the optical enhancement structure; etching the first clad layer to form a plurality of bioreaction chambers spaced apart along a top surface of the first clad layer; depositing a receptor layer over the first clad layer and lining the first clad layer at the plurality of bioreaction chambers; bonding a capping layer over the first clad layer and over the receptor layer, wherein a microfluidic channel extends through the capping layer along a top surface of the receptor layer and along the plurality of bioreaction chambers; and bonding the dielectric structure, first clad layer, receptor layer, and capping layer over the plurality of photodetectors, wherein the optical signal enhancement structure is between the plurality of bioreaction chambers and the plurality of photodetectors and configured to enhance an optical signal emitted from bioreactions occurring along the plurality of bioreaction chambers.
7. The method claim 6, wherein forming the optical signal enhancement structure comprises: forming an incident radiation filter layer on a bottom surface of the dielectric structure so that incident radiation filter layer is between the dielectric structure and the micro-lenses.
8. The method of claim 7, wherein forming the optical signal enhancement structure further comprises: depositing an aperture layer along a bottom of the incident radiation filter layer; etching the aperture layer to form a plurality of first apertures in the aperture layer; and depositing a glue layer over the plurality of micro-lenses so that the glue layer is between the aperture layer and the plurality of micro-lenses, wherein portions of the glue layer fill the plurality of first apertures, and wherein the portions of the glue layer are directly under the plurality of bioreaction chambers.
9. The method of claim 8, wherein forming the optical signal enhancement structure further comprises: forming a focusing layer within the dielectric structure; and etching the focusing layer to form a plurality of focusing segments from the focusing layer, the plural of focusing segments directly below the plurality of bioreaction chambers, respectively.
10. The method of claim 9, wherein forming the optical signal enhancement structure further comprises: depositing a second clad layer over the dielectric structure, wherein the second clad layer is between the first clad layer and the dielectric structure; and depositing a waveguide channel core layer over the second clad layer, wherein the waveguide channel core layer is between the first clad layer and the second clad layer, wherein the bioreaction chambers are directly over a top surface of the waveguide channel core layer.
11. The method of claim 10, further comprising: depositing a reflector layer over the focusing layer, wherein the reflector layer is disposed along a top surface of the dielectric structure and a bottom surface of the second clad layer, wherein the reflector layer has a plurality of second apertures therein, wherein the second clad layer is over the reflector layer, wherein portions of the second clad layer fill the plurality of second apertures, and wherein the portions of the second clad layer are directly under the plurality of bioreaction chambers.
12. A method for forming an integrated chip, the method comprising: forming a plurality of photodetectors along a semiconductor layer; forming a plurality of color filters over the plurality of photodetectors, respectively; forming a plurality of micro-lenses over the plurality of color filters,

respectively; depositing a glue layer over the plurality of micro-lenses; forming an optical signal enhancement structure along a dielectric structure, wherein forming the optical signal enhancement structure comprises forming a radiation filter layer under the dielectric structure; forming a plurality of bioreaction chambers over the dielectric structure and over the optical signal enhancement structure; depositing a receptor layer along the plurality of bioreaction chambers; bonding a capping layer having a microfluidic channel therein over the receptor layer and over the plurality of bioreaction chambers; and bonding the radiation filter layer and the dielectric structure over the plurality of photodetectors and along the glue layer such that the plurality of bioreaction chambers are disposed over the plurality of photodetectors.

13. The method of claim 12, wherein forming the optical signal enhancement structure comprises forming an aperture layer on the radiation filter layer.

14. The method of claim 12, wherein forming the optical signal enhancement structure comprises forming a plurality of focusing layer segments within the dielectric structure.

15. The method of claim 12, wherein forming the optical signal enhancement structure comprises forming a reflector layer over the dielectric structure.

16. The method of claim 12, further comprising: forming a temperature sensor within the dielectric structure.

17. The method of claim 12, further comprising: forming a heater within the dielectric structure.

18. The method of claim 12, further comprising: forming a waveguide layer over the dielectric structure and over the optical signal enhancement structure, wherein the plurality of bioreaction chambers are formed along the waveguide layer, and wherein the receptor layer is deposited on the waveguide layer.

19. The method of claim 18, further comprising: forming a light coupler structure along the waveguide layer.

20. The method of claim 12, wherein the microfluidic channel extends through the capping layer along the receptor layer and the plurality of bioreaction chambers.